

**Les Clewlow** and **Chris Strickland** are the founding directors of the Lacima Group through which they provide banking, weather and energy derivatives consulting, software and training services. They are the authors of the book "Implementing Derivatives Models" (Wiley, 1998) and editors of "Exotic Options: The State of the Art" (ITP, 1998) and have published many articles in both academic and trade journals. They also hold Associate Research Positions at the University of Technology, Sydney and the University of Warwick, UK. Les and Chris received their Ph.D.s in Computer Science and Finance, respectively, from the University of Warwick.

**Vincent (Vince) Kaminski** is Managing Director and Head of Research for Enron Corp., which he joined in June 1992. Previously, Vince was a Vice President in the research department of Salomon Brothers in New York and a manager in AT&T Communications in Bedminster, New Jersey. Vince is a recipient of the 1999 James H. McGraw Award for Energy Risk Management (Energy Risk Manager of the Year). Vince holds an M.S. degree in international economics and a Ph.D. degree in mathematical economics from the Main School of Planning and Statistics in Warsaw, Poland, and an MBA from Fordham University in New York.

**Grant Masson** is a vice president with the research group of Enron Corp. He oversees the quantitative support for asset and derivative structure valuations and market analysis for both domestic and international electricity trading and origination. Prior to joining Enron in 1994, Grant spent five years at the University of Basel as a research scientist specializing in experimental nuclear physics. He received a BA from Rice University in Houston and MS and Ph.D. degrees in physics from the University of Wisconsin - Madison.

**Ronnie Chahal** is a manager in the Enron Research group. He currently provides analytical support to the risk management group in Enron Energy Services. He has a Ph.D. in Finance from Georgia State University, Atlanta, Georgia.

# ENERGY DERIVATIVES PRICING AND RISK MANAGEMENT

Les Clewlow and Chris Strickland

With contributions and published in association with Enron Corp.

Energy markets around the world are rapidly being deregulated leading to unprecedented levels of competition in the energy industry, increased exposure to the prices on energies, and exposing participants to potentially catastrophic risks. This book provides a comprehensive and technical treatment of the valuation and risk management of energy derivatives, within the oil, gas, and electricity markets.

*"...a thorough and comprehensive overview of energy derivatives markets, instruments, derivatives pricing and risk management methodologies...This is the first book that clearly explains, in detail, energy price behavior and the different modeling approaches employed to capture the complex realities of these markets. This book should become a standard reference for energy derivatives traders, risk managers, energy risk consultants, and anyone else interested in the fascinating area of energy derivatives."* — Mark Garman, Professor Emeritus of Finance, Walter A. Haas School of Business Administration, at University of California, Berkeley, CEO of Financial Engineering Associates, Inc.

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This book provides a comprehensive and technical treatment of the valuation and risk management of energy derivatives, within the oil, gas, and electricity markets, and looks in depth at:

- the behaviour of spot and forward prices
- modelling and estimating volatility
- structures that exist in the energy derivatives market and their applications
- pricing and hedging of energy derivatives using single and multi-factor spot price models
- using simulation and trinomial trees for pricing caps, floors, swaptions, and path-dependent securities such as barriers, Asians, lookbacks, and swing options.
- multi-factor models of the forward curve, including volatility estimation and pricing of exotic derivatives
- risk management of energy derivative positions
- value-at-risk and credit risk for energy derivatives.

A large proportion of the content is original research by the authors who have applied over 20 years combined derivatives experience and research to the energy markets.

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